

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.2: Chemicals of Life — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Driving Question	DRIVING QUESTION: How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 A Plain Plate with Extraordinary Results	What did we observe? Kipchoge ran 42.195 km in 7,269 s, a pace of 172.3 s per km, and the marathon costs about 2,200 kcal while stored glycogen is about 2,000 kcal. Table 1 shows that ugali, githeri and beans contain carbohydrate, protein and a little fat.	What did we learn? Carbohydrates, proteins and lipids are the three main groups of chemicals of life. Carbohydrates supply and store energy, proteins build and repair the body, and lipids store a lot of energy and insulate. Water, vitamins, minerals and nucleic acids are also chemicals of life, which we will meet later.	How does this help explain the phenomenon? A plate of ugali, beans, sukuma wiki and milk is not plain: it contains the chemicals of life that the body needs for energy, growth and repair. We still need to find out how we can show these chemicals are in the food and how the body uses them, so we can explain how simple foods fuel a marathon runner from Iten.
Lesson 2 Carbohydrates: starch, sugars and how to test for them	We observed that maize flour and sweet potato turned iodine blue-black, ripe banana gave only a faint brown-orange with iodine, and Benedict's gave blue for maize flour, green-yellow for sweet potato, orange to brick-red for ripe banana, and blue for sugarcane juice until it was boiled with dilute HCl and then gave brick-red.	We learned that carbohydrates include monosaccharides (glucose, fructose), disaccharides (sucrose, maltose) and polysaccharides (starch, glycogen, cellulose); iodine turns blue-black with starch; Benedict's detects reducing sugars (glucose, fructose, maltose) but not the non-reducing sugar sucrose until it is split by boiling with dilute HCl. Table 1 shows 26 g carbohydrate per 100 g in ugali, 23 g in ripe banana, 20 g in boiled sweet potato and 18 g in sugarcane juice.	We can explain that ugali is not plain: its maize starch is a long chain of glucose units that stores energy, and the tests give evidence for this. This helps explain how simple Kenyan foods supply the carbohydrate energy that athletes like Eliud Kipchoge need, but we still need to find out how the body breaks starch down and what proteins and lipids do.
Lesson 3 Proteins: building and repairing the body	We tested beans, milk, egg white, omena and ugali with Biuret reagent. Beans, milk, egg white and omena turned violet; ugali	Proteins are long chains of amino acids. Biuret turns blue to violet when protein is present. Foods like beans, milk, egg and	Athletes in Iten eat beans, milk and omena with ugali because the proteins are digested to amino acids, which the body uses to

	stayed blue. Table 1 shows omena has the most protein (60 g per 100 g), then egg (13 g), beans (8 g), milk (3.3 g) and ugali (2.5 g).	omena are good protein sources, while ugali has too little to detect.	grow and repair muscle damaged by running, while ugali supplies starch for energy. This adds protein to our model of how a plain plate supports Kipchoge's performance.
Lesson 4 Lipids: stored energy and insulation	Groundnuts, cooking oil, mandazi and nyama choma left a translucent spot on brown paper that stayed after drying, and groundnuts and cooking oil gave a cloudy white layer in the emulsion test. Maize flour left no translucent spot. Table 1 shows groundnuts have 49 g fat per 100 g, mandazi 15 g, nyama choma 15 g and ugali only 0.6 g.	Lipids (fats, oils and waxes) are made of fatty acids and glycerol. Lipids give 9 kcal per g, more than twice the 4 kcal per g of carbohydrate. They store energy and insulate the body.	Groundnuts have 609 kcal per 100 g (64 + 104 + 441) compared with 119.4 kcal for ugali, because fat holds 9 kcal per g. Lipids in foods like milk, groundnuts and cooking oil add stored energy and insulation to a runner's body, but the main fuel for the marathon (about 2,200 kcal) is carbohydrate stored as glycogen (about 2,000 kcal).
Lesson 5 Enzymes: Why Chewed Ugali Tastes Sweet	Chewed ugali tastes sweeter. In the investigation, starch with saliva amylase cleared fastest at 37 C (60 s, rate 1.0) and at pH 6.8 (60 s, rate 1.0). At 20 C it took 300 s (rate 0.2). At 0, 60 and 80 C and at pH 2 and 12, starch was not cleared in 600 s (rate 0). The Benedict's test showed a reducing sugar after amylase acted.	Enzymes are protein catalysts made of amino acids. Amylase breaks starch into maltose, a reducing sugar. Enzymes work best at an optimum temperature (about 37 C) and pH (near neutral). Extreme heat or pH denatures an enzyme permanently. Each enzyme is specific to its substrate.	Ugali tastes sweet when chewed because saliva amylase, a protein enzyme, breaks starch into maltose at body temperature. The reaction at 37 C is 5 times faster than at 20 C (1.0 / 0.2). Without enzymes working at their optimum, the starch in ugali could not be turned into glucose quickly enough to fuel a runner like Kipchoge from Iten.
Lesson 6 Which meal best supports the athlete?	We compared Meals A, B and C using Table 4: Meal A gave 763.7 kcal with 126.25 g carbohydrate, 30.25 g protein and 18% of energy from fat; Meal B gave 477 kcal; Meal C gave 921 kcal but 44% of its energy came from fat and only 90 g carbohydrate. We compared Kipchoge's marathon cost of about 2,200 kcal with his glycogen store of about 2,000 kcal, and we compared our Lesson 1 model with our final model.	Ugali, beans, sukuma wiki and milk contain all three chemicals of life: carbohydrates (starch) for energy, proteins (amino acids) for growth and repair and for making enzymes, and lipids for stored energy and insulation. Enzymes such as amylase work best at about 37 C and near neutral pH and break big food molecules into small ones the body can absorb. $\text{Energy} = 4 \times \text{C} + 4 \times \text{P} + 9 \times \text{F}$ lets us compare meals with numbers.	A plain plate of ugali and beans is not plain: it supplies carbohydrates that enzymes break down to glucose for energy and glycogen storage, proteins that supply amino acids for repair and growth, and some lipids for stored energy. Meal A (ugali, beans, sukuma wiki and milk) best supports an endurance athlete because it gives the most carbohydrate (126.25 g), adequate protein (30.25 g) and moderate fat (18% of energy). This is how simple Kenyan foods can fuel a 2:01:09 marathon like Eliud Kipchoge's.

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